s Used AI

and How We Hope It Will Be Used in the Future

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List of Slides (DO NOT DELETE)

(Slide 2) Terminology Review-Jing

NLP, Training Dataset, supervised/unsupervised (refer to Module 1)

(Slide 3-4) Atticus overview-Jing

What is Atticus working on?

Training Dataset

not product or code/AI model

How does Atticus do it?

Use eBrevia to annotate/label selected clauses [Link to Published Labels on Atticus Website)

Labeling inside peek - Screenshots from eBrevia

Science behind atticus

1. More documents, higher accuracy (Anya)

2. Should be a # of documents where it will guarantee a good f-score (0.80) (Andrew)

3. We can find out when AI reaches a consistent accuracy and reallocate human labelling (based on f-score) (Tagir)

We can look at multiple algorithms and find out which ones are generally easy or hard

Different algorithms will produce different results

Concept Review

Review what was covered in module 1

Format based on modules 2 and 3

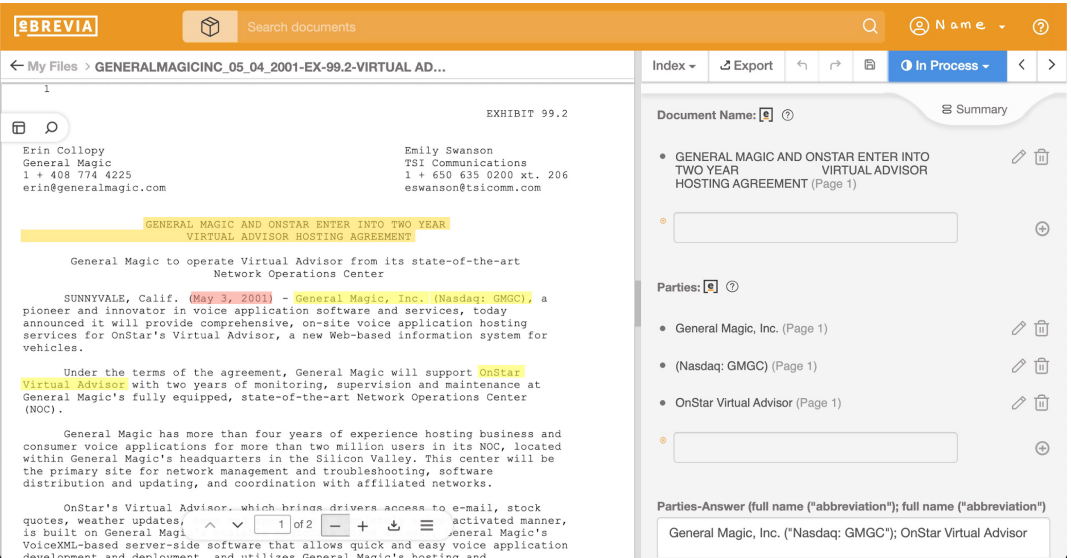
Atticus Overview

What is Atticus working on?

We are training a dataset, not making a product

How does Atticus do it?

We use eBrevia to annotate and label certain clauses



Here’s a sneak peek inside eBrevia!

Science Behind Atticus - Hypothesis 1

Hypothesis: More examples = higher f-score/accuracy

Hypothesis 1: More Examples = Higher Accuracy

Purpose:

Hypothesis 1: More Examples = Higher Accuracy

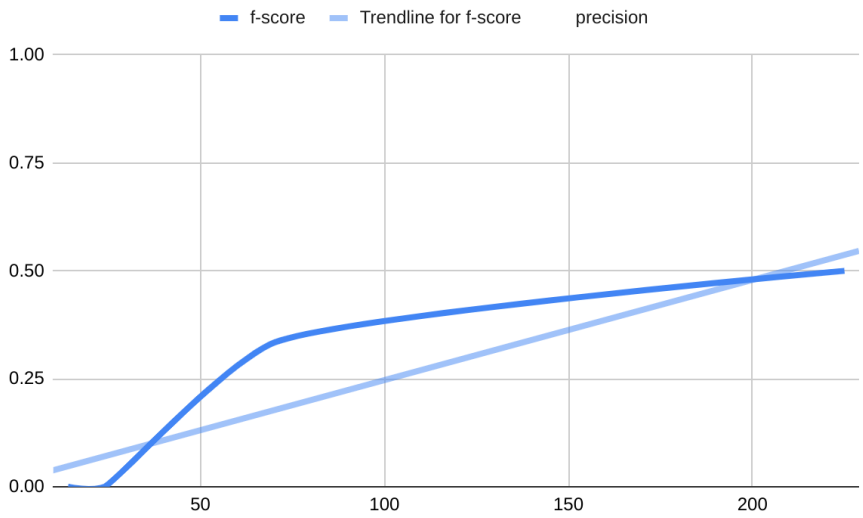
Background / Research:

Hypothesis 1: More Examples = Higher Accuracy

Procedures:

Checked automatic trendline created by Google Sheets.

Differences that either I didn't notice, or ones with net difference equal to or less than .01, were considered "same".



Hypothesis 1: More Examples = Higher Accuracy

Data

Hypothesis 1: More Examples = Higher Accuracy

Results:

True for 17/31 of labels

Same for 9/31 of labels

False for 5/31 of labels

So, only true for an iffy 55% of labels

Hypothesis 1: More Examples = Higher Accuracy

Conclusion:

Only true for a bit more than half of them,

(looking into the no's rn)

Why is this useful/important/significant?

Things learned, learnings will be used for?

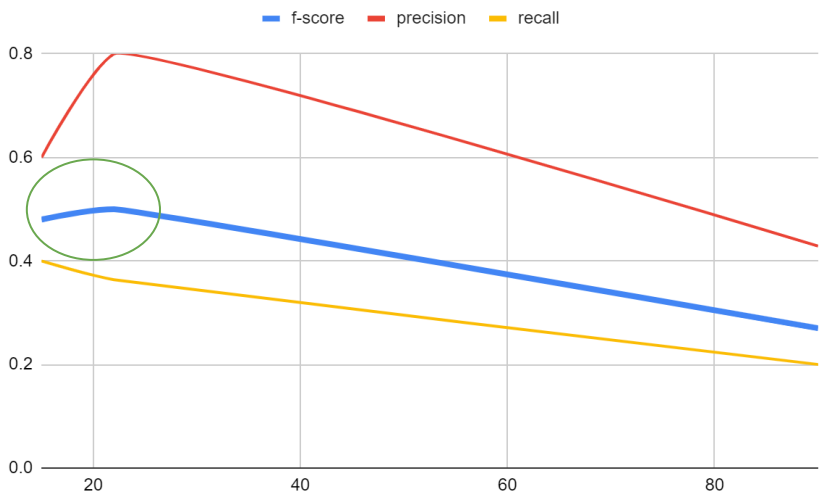
Science Behind Atticus - Hypothesis 2

Hypothesis 2: There is usually a range of number of examples where the best f-scores (hopefully 0.80+) can be reached.

Background and Research

What we are trying to do in this experiment is to prove that the AI will behave similarly if we give it different models with similar numbers of examples

Some of the labels never reach a good f-score, but that doesn't matter for this hypothesis.



This is taken from the label "Non-Compete". The green circle marks the sweet spot, where the highest f-scores are seen. Even though they aren't high, it doesn't matter for this hypothesis.

Procedures

If two or more different models with similar number of examples (at most 10 examples apart) yield consistent results (at most 0.05 apart), the label goes into the “consistent” section

Only applies if both f-scores are among the highest. If they are, they are colored green. Otherwise, they are colored orange.

If all models have an f-score of 0 the label goes into the “unreliable data” section

Colored red

Otherwise, they go into the “tentative” section. The reason is that since there is always one model with the highest score, another future model with a similar number of examples can yield a similarly high score.

If a label had no two models that were less than 10 examples apart, it gets colored blue

Data

Results

Requirements met for 9 labels

7 colored green

2 colored orange

Requirements yet to be met for 20 labels

16 colored blue

4 colored red

AI consistently fails for 3 labels

Total: 32 labels

Conclusion

While most of the labels have yet to satisfy these conditions, we have located the “sweet spot” for some of these already.

When we test more and more, we will probably find more of these “sweet spots”.

Science Behind Atticus - Hypothesis 3

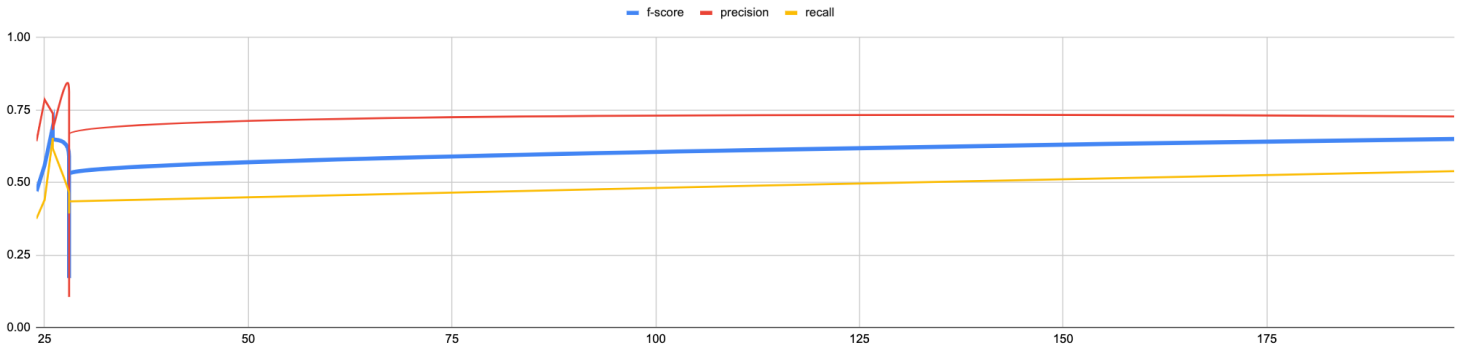
We can find out when AI reaches a consistent accuracy and reallocate human labelling (based on f-score)

Background / Research

Based on Hypothesis one, if there are more examples, most of the labels will have a higher F-Score.

Some labels have enough examples, that the F-Score will stabilize, and change more slowly if you keep adding more contract examples.

As shown in the chart below, the more examples you have for signing date, the more the F-Score Increases, however, it becomes more consistent and stagnant.



Procedures

If Label has an f-score higher than 0.6 and is consistent, it will go in the consistent label section.

If Label is either significantly changing in f-score and has at least 3 models with at least 40 examples in the last model, it is in the inconsistent label section.

If Label has either less than 3 models, less than 40 examples on the last model, or is consistent at a low score (lower than 0.6 f-score), it will go in the Not Enough Models and Examples section.

Not Enough Models and Examples Color Code for Labels:

Green = Not Enough Models (but enough Examples). Purple = Red and Blue.

Blue = Not Enough Examples and/or Models.

Red = Inconsistent at a low F-Score.

Data

Results

4 Consistent Labels

7 Inconsistent Labels

21 Not Enough Examples/Not Enough Models/ Consistent low F-Score.

32 Labels Total

Colored Labels

6 Green Labels

3 Blue Labels

3 Red Labels

9 Purple Labels

Conclusion

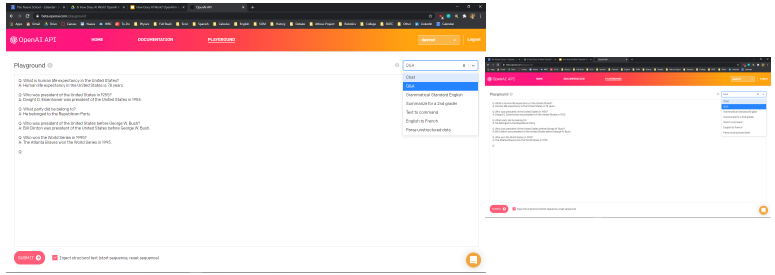
There are a lot of Labels that need more examples and/or models, it is best to focus human time on those.

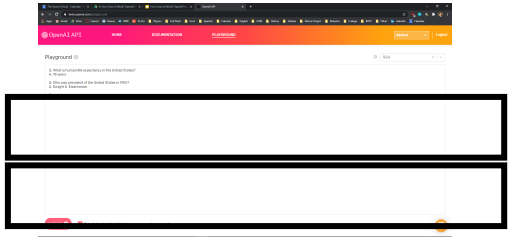
Also, there were labels that were consistent at low F-Scores which is most likely due to the NLP system itself. It is better to not focus on those since it is a problem with eBrevia rather than the dataset itself.

Ease of Use: Playground

GPT-3, currently in beta, is available through an online playground with presets such as a chatbot, Q&A, and “summarize for a 2nd grader.”

In addition, users can give the AI example question and answer pairs to teach it the structure they want the output to look like. The model can learn from just one or two examples.

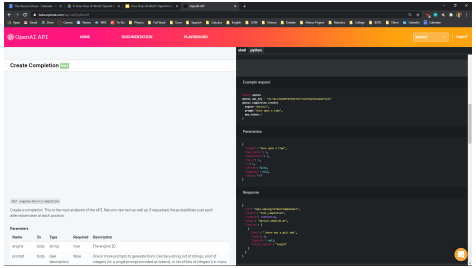
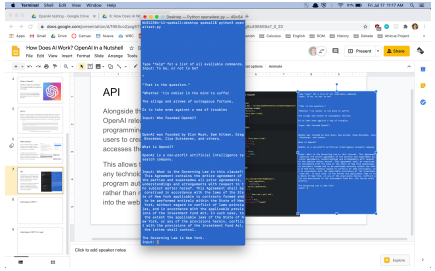




Sample structure provided by user

Subsequent output matches the structure

Ease of Use: API

Alongside the web interface for GPT-3, OpenAI released an API (application programming interface), which allows users to create their own code that accesses the AI model.

This allows the model to be applied to any technology and for users to program automated Q&A systems rather than manually typing questions into the web interface.